



Forsta Open Dialer Interface

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Definitions

The following terms are used in this document (they are marked in **Bold** throughout the document).

Agent – a person working with the dialer as an interviewer

Agent group – a group of agents which has its own identifier

Campaign – corresponds to Forsta CATI survey

Forsta CATI – an add-on to Forsta which provides CATI (Computer Assisted Telephone Interview) functionality

Forsta open dialer interface assembly – an assembly (dll file) supplied by Forsta to **Dialer providers**. This assembly contains definitions of all common constants, enumerations, classes, exceptions, interfaces.

Forsta dialer WS (Forsta dialer web service) – **Forsta CATI** interacts with the **dialer** through this web service.

Forsta open dialer interface – a set of interfaces and classes providing interaction between **Forsta CATI** and the **Dialer provider**

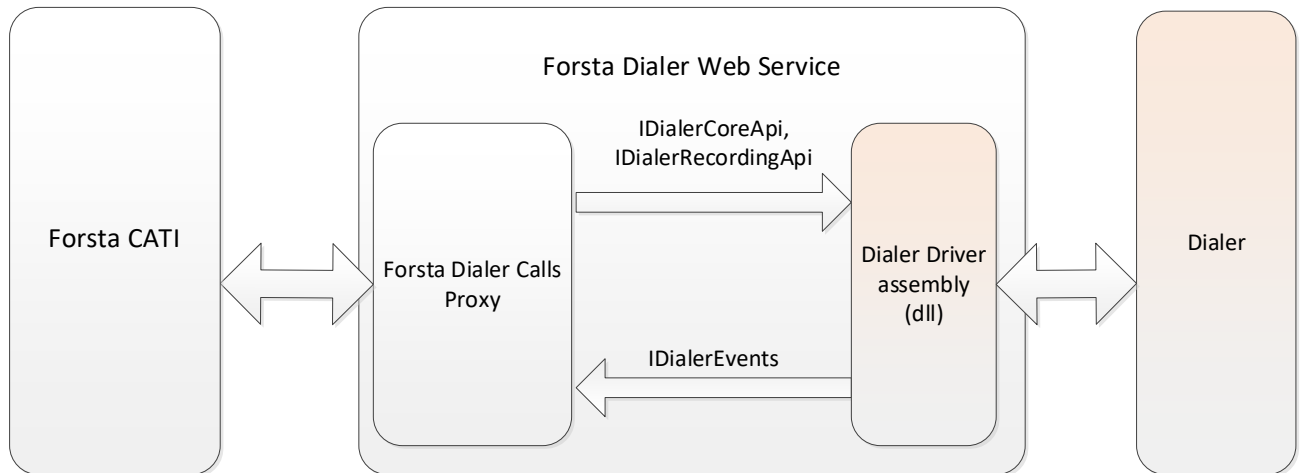
Dialer – (in this document) a dialer provided by the **Dialer provider**

Dialer driver assembly – an assembly (dll file) provided by the **Dialer provider** to Forsta. This assembly implements **Forsta open dialer Interface** and function inside **Forsta dialer WS**.

Dialer provider – a company which uses **Forsta open dialer interface** to attach its own **dialer** to **Forsta CATI**

Common schema or How does it work

Forsta CATI interacts with the **dialer** via **Forsta dialer web service**.



What Forsta CATI must supply to the dialer provider

Forsta CATI provides:

1. Public **Forsta open dialer interface assembly** <ConfermitDialerInterface.dll> which contains definitions of all interfaces and classes required for **Forsta CATI** <-> **Dialer** interaction.

What the dialer provider must supply

In order to integrate a dialer into **Forsta CATI** the **dialer provider** should supply:

1. **Dialer driver assembly** (example: ProviderDialer.dll) which implements **Forsta open dialer Interface**
2. **Dialer driver assembly** main class name and name space
Example:
Namespace "NewDialer.Dialer"
Main class "NewDialerLibrary"
(This information allows **Forsta dialer WS** to create an instance of the Dialer driver object.)

3. A file in XML format containing the dialer parameters

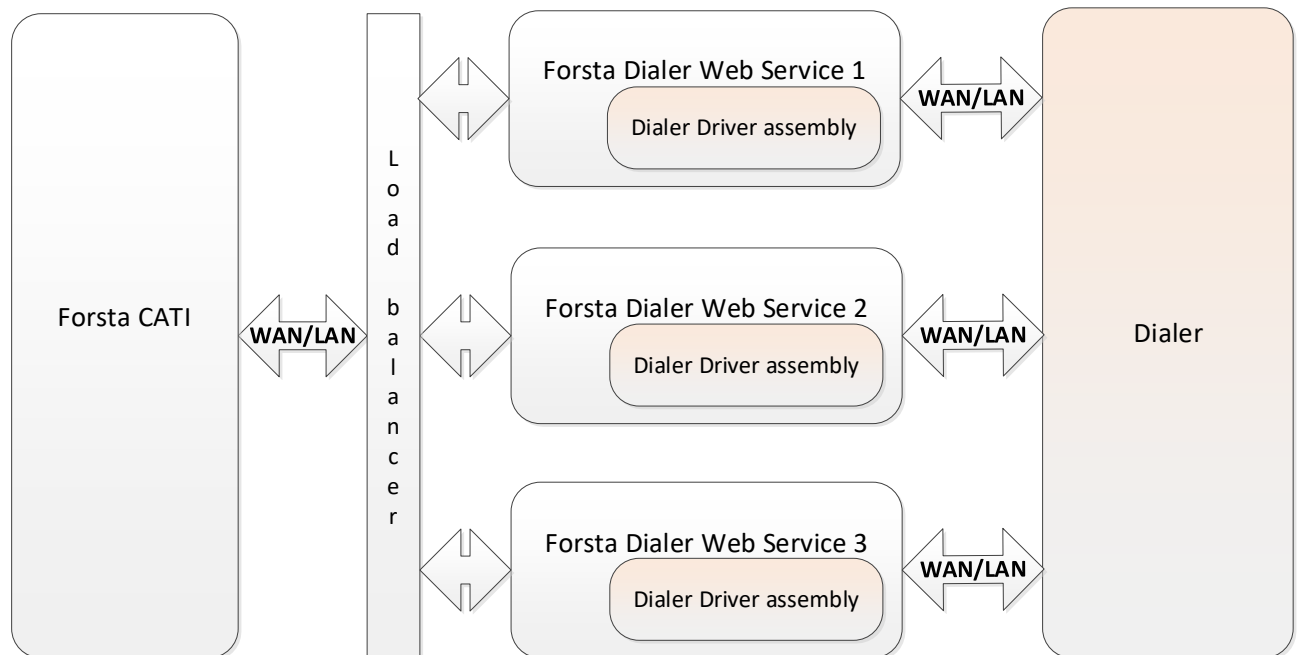
- Dialer configuration parameters
- Dialer default campaign parameters

See [Dialer parameters](#) for details.

Scalability

If the **Dialer** is able to work with a large amount of interviewers simultaneously then **Dialer provider** should pay special attention to the stateless implementation of the **Dialer driver assembly**.

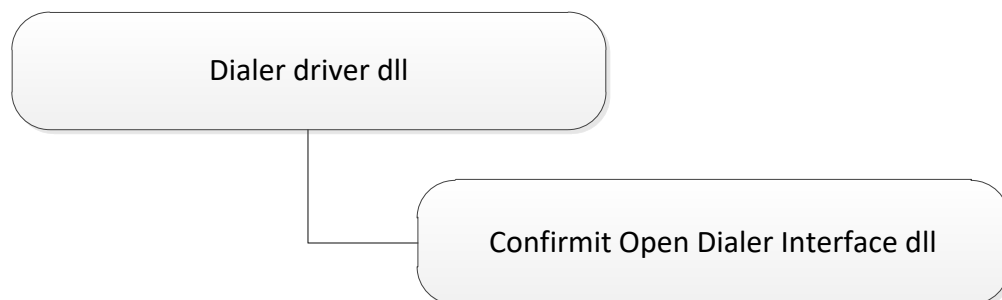
Forsta CATI uses a number of **Forsta dialer Web Services** running in parallel when it works with powerful dialers. Note that commands and notifications can be routed through different servers, for example dial command can go via server 1 and response outcome notification can go via server 2.



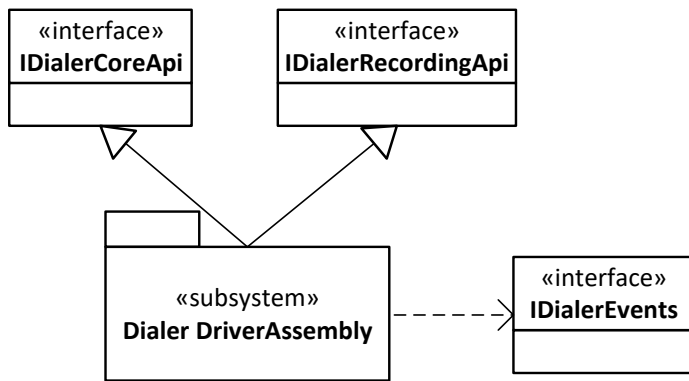
Dialer driver assembly implementation

Forsta provides the **Forsta open dialer interface assembly**.

To access all interaction interfaces and common classes the **Dialer driver assembly** must include reference to **Forsta open dialer interface assembly**:

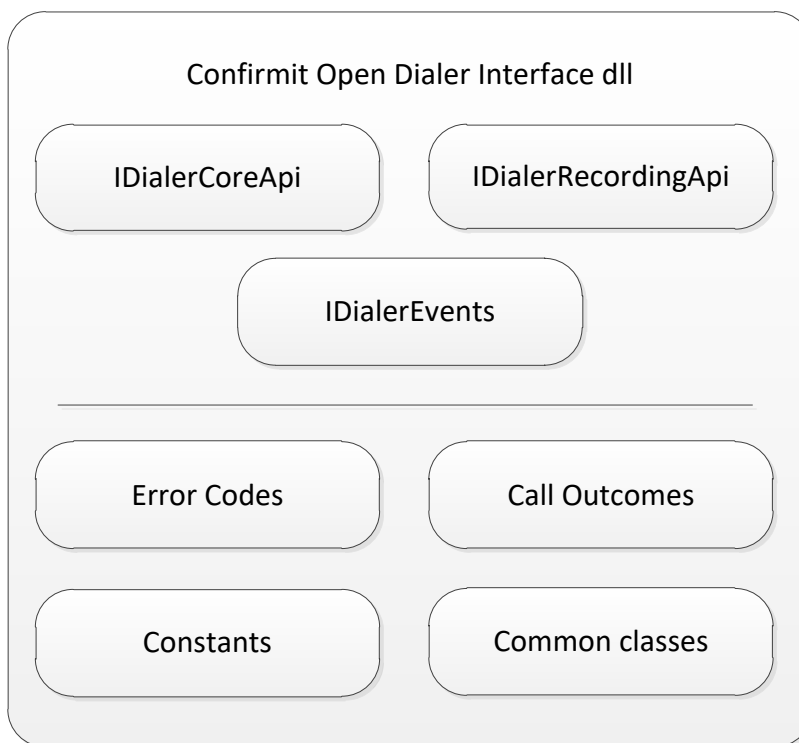


Dialer driver assembly implements **IDialerCore** and **IDialerRecording** interfaces and uses **IDialerEvents** interface for notifications directed from the **dialer** to CATI:



Forsta Open dialer interface assembly structure

The schema below shows the contents of the **Forsta open dialer interface assembly**.



Details regarding all **Open dialer interface assembly** elements are available in a separate help file.

Workflow

Instantiation

First **Forsta dialer WS** creates an instance of the Dialer driver object. It always uses a constructor with the next signature:

```
public SomeDialerDriver(IDialerEvents dialerEvents, ILogger logger);
```

Parameters:

dialerEvents - **Dialer** uses this reference to send notifications to **Forsta CATI**

logger - **Dialer** can use this logger to log messages into **Forsta CATI** log.

So **Dialer driver assembly** main class must have a constructor with such signature.

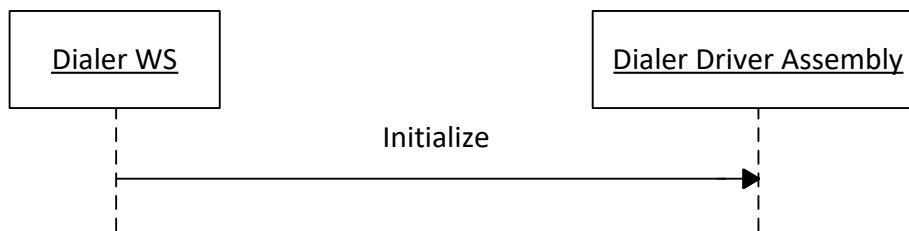
Initialization

Dialer driver assembly provides two interfaces:

- **IDialerCore**
- **IDialerRecording**

For each interface **Forsta CATI** first calls the **Initialize** method. Then it can call other interface methods.

Forsta CATI can call the **Initialize** method several times; it does call the **Initialize** method each time **Confrmit CATI** backend service starts.



There is a set of **IDialerCore** methods that **Forsta CATI** can call at any time, even before **Initialize**. These methods are used to configure common properties of the dialer:

string GetName() – returns **Dialer** name in order to show it in CATI supervisor. Ex.: **TCL**.

string GetVersion() – returns **Dialer** version to show it in CATI supervisor. Ex.: **1.0**.

ValidateConfigurationParameters(string parametersXml) – checks that **parametersXml** contains correct dialer configuration parameters.

SetConfigurationParameters(string parametersXml) – configures the dialer with the new parameters.

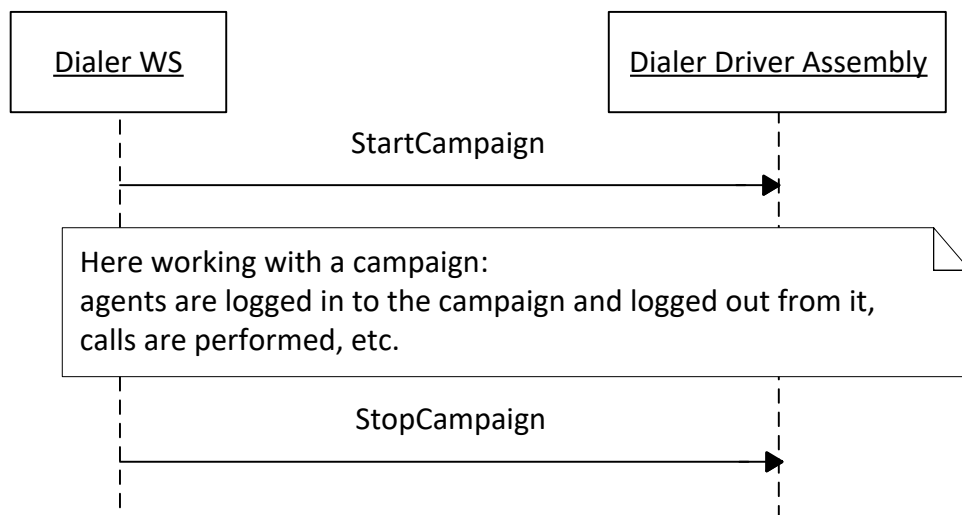
StartCampaign / Stop campaign

Forsta CATI sends StartCampaign command when the corresponding survey is opened in **Forsta**.

StartCampaign is a synchronous method. It means that **Forsta CATI** does not expect any asynchronous notification from **Dialer**, as soon as the method returns successfully the corresponding campaign is considered by **Forsta CATI** as opened.

Forsta CATI sends StopCampaign command when the corresponding survey is closed in **Forsta**.

StopCampaign is a synchronous method as well.



Note

Dialer must not immediately logout agents working with the campaign when the campaign is stopped. It must allow the logged in agents to finish their current calls and it must stop delivering new calls for the campaigns.

Agent login / Agent logout and agent ready status

Depending on the agent task choice in Forsta CATI agent can login either to a particular campaign or not. Login and Logout operations are asynchronous, **Dialer** informs **Forsta CATI** that login or logout operation is finished via **IDialerEvents.NotifyAgentState** method.

As soon as an agent is logged in **Forsta CATI** calls GoReady method to make the agent available for calls.

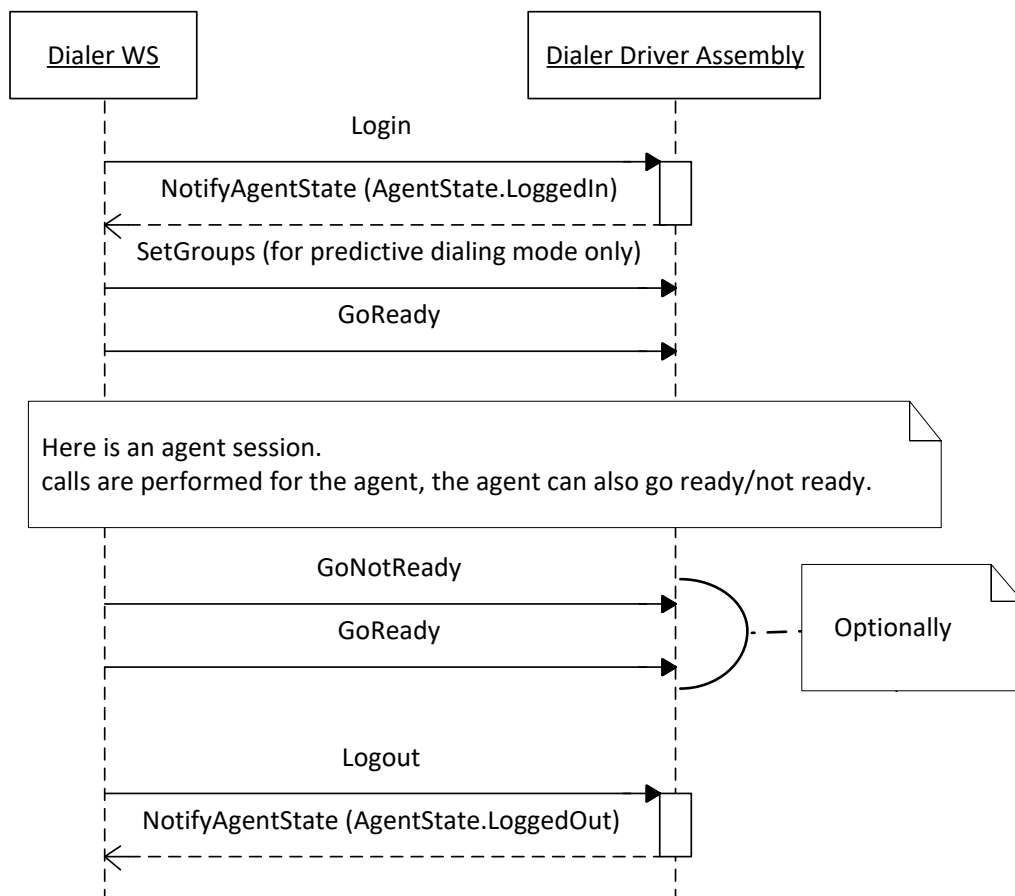
Note

If agent is logged in in predictive dialing mode then **Forsta CATI** calls SetGroups method just before GoReady to tell **Dialer** what agent groups the agent belongs to.

During the agent session Forsta CATI can call GoNotReady to make the agent not available for calls, and GoReady to return the agent back to working state. GoReady and GoNotReady operations are synchronous.

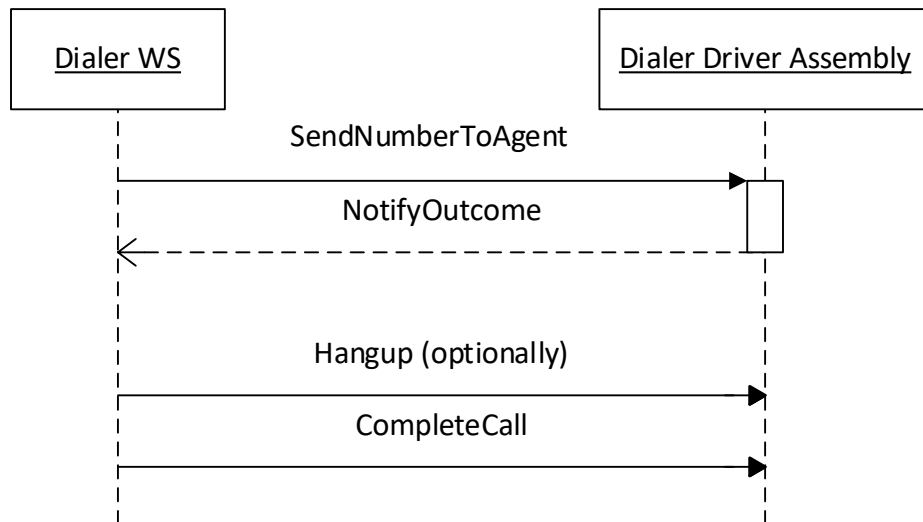
Note

Forsta CATI calls GoNotReady only when the agent is not on a call. **Forsta CATI** can also make agent not ready at CompleteCall operation (setting makeAgentReady parameter to false)



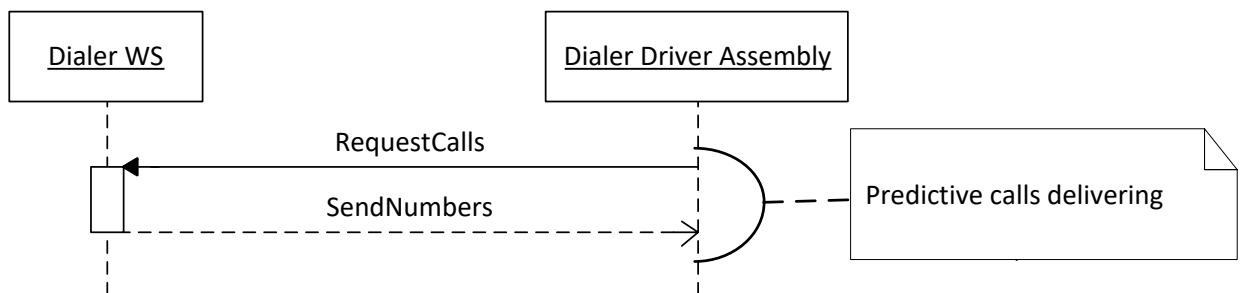
Interview cycle in automatic/preview dialing modes

Interview cycle in automatic and preview dialing modes looks like as follows:



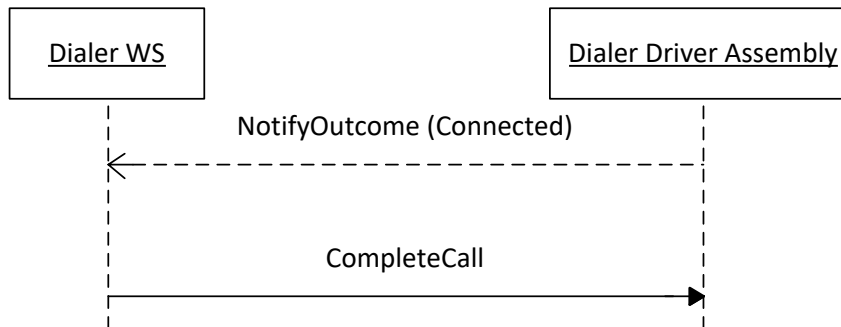
Interview cycle in predictive dialing mode

As soon as a campaign is started in predictive dial mode **Dialer** can request **Forsta CATI** for calls, **Dialer** calls RequestCalls each time it needs more calls, **Forsta CATI** answers each request with SendNumbers. If there are no calls to dial SendNumbers method contains empty call list.



Dialer tries to call the numbers, when a call is connected **Dialer** assigned the call to an agent and

interview starts. When interview is finished Forsta CATI calls CompleteCall:

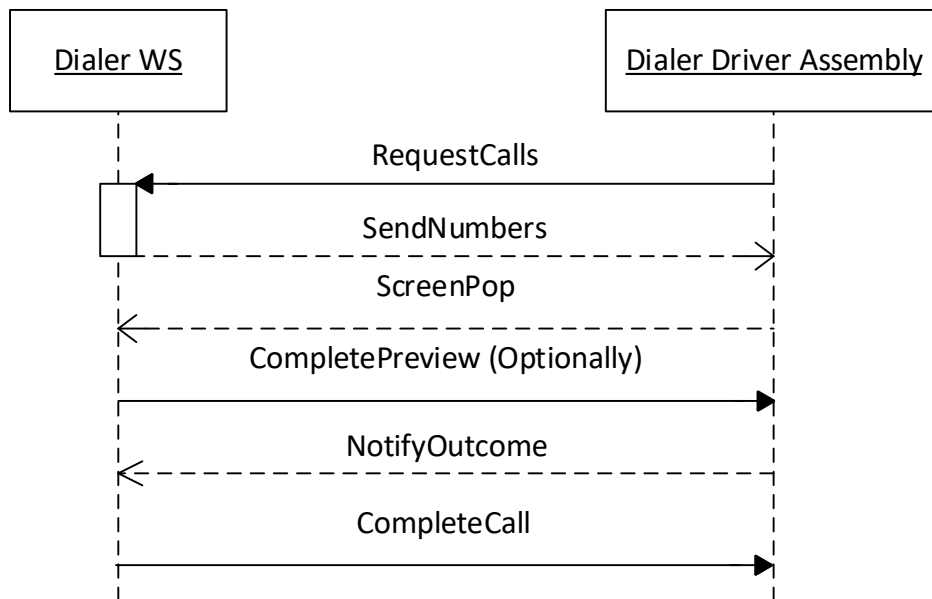


Hybrid dialing in predictive dialing mode

While a **campaign** is opened in predictive dialing mode some calls can be dialed in preview dialing mode. Such behavior is called **hybrid dialing**, and such calls are called **hybrid calls**. Here are the hybrid dialing steps:

- **Dialer** requests calls via **IDialerEvents.RequestCalls** method.
- **Forsta CATI** answers via **IDialerCoreApi.SendNumbers**, some calls in the call list can have dialing mode = **DialingMode.Preview**, such calls are to be considered by dialer as hybrid calls.
- **Dialer** does not immediately dial hybrid calls. Instead of it **dialer** waits for any available **agent** for this call, and as soon as the **agent** is found dialer “attaches” the **agent** to the call and notifies **Forsta CATI** that call is ready for dialing via **IDialerEvents.ScreenPop** message.
- Now **Forsta CATI** can make real call to the respondent via **IDialerCoreApi.CompletePreview** command. But **Forsta CATI** is not obliged to call **CompletePreview**. It can miss this step.
- **Forsta CATI** sends **IDialerCoreApi.CompleteCall** to complete the call and release the agent for the next calls.
- Later on **Forsta CATI** sends **IDialerCoreApi.UpdateInterviewStatus** to inform **Dialer** about the final interview status.

And here is the hybrid dialing workflow diagram:



Dialer health control

To ensure that the dialer is operational **Forsta CATI** repeatedly calls **IDialerCoreApi.GetState**.

When a value *DialerState.Available* is returned, it indicates that the dialer is “alive”.

If the dialer returns *DialerState.Unavailable* or the **GetState** call fails then Forsta CATI tries to call **GetState** again and again during specified time interval. If **GetState** call still returns *DialerState.Unavailable* after the specified time interval expires, Forsta CATI assigns dialer the “unavailable” state and calls **IDialercoreApi.Release**.

The **dialer** can also use **IDialerEvents.NotifyDialerState** to notify Forsta CATI about dialer state change.

Dialer parameters

Forsta CATI stores two types of dialer parameters in its own database:

- Dialer configuration parameters
- Dialer campaign parameters

The following **IDialerCoreApi** functions work with dialer parameters:

Dialer configuration parameters:

SetConfigurationParameters(string companyId, string configurationParametersXml)

Dialer campaign parameters:

SetCampaignParameters(
string companyId,
int campaignId,
DiallingMode dialingMode,
string campaignParametersXml)

The **Dialer provider** supplies an XML file containing the default configuration and campaign parameters in the following format:

<DialerConfigurationParameters>

 <DialerParameter>

 <Id>... </Id>

 <Name>... </Name>

 <Value>...</Value>

 <Type>... </Type> - A .Net simple type is required here: String, Boolean, Int32.

 <MinValue>...</MinValue> - if applicable

 <MaxValue>...</MaxValue> - if applicable

 <PossibleValues>...</PossibleValues> - if applicable

 <Visible/>...</Visible> - True or False. True means the parameter is visible in CATI

Supervisor

 <Editable/>...</Editable> - True or False. True means the CATI supervisor can edit the parameter value.

```
</DialerParameter>

<DialerParameter>
    ...
</DialerParameter>
</DialerConfigurationParameters>
<DialerCampaignParameters>
    <DialerParameter>
        <Id>... </Id>
        <Name>... </Name>
        <Value>...</Value>
        <Type>... </Type>
        <MinValue>...</MinValue>
        <MaxValue>...</MaxValue>
        <PossibleValues>...</PossibleValues >
        <Visible>...</Visible>
        <Editable >...</Editable >
    </DialerParameter>
    <DialerParameter>
        ...
    </DialerParameter>
</DialerCampaignParameters>
```

Example:

```
<DialerParameter>
    <Id>MaxRings</Id>
    <Name>No reply timeout (no. of rings) </Name>
    <Value>5</Value>
```

```
<Type>System.Int32</Type>

<MinValue>0</MinValue>

<MaxValue>10</MaxValue>

<Visible>True</Visible>

<Editable >True</ Editable>

</DialerParameter>

<DialerParameter>

    <Id>AnsMachineDetect</Id>

    <Name>Enable answer phone detection</Name>

    <Value>False</Value>

    <Type>System.Boolean</Type>

    <Visible>True</Visible>

    <Editable >True</ Editable>

</DialerParameter>
```

Forsta CATI validates parameters according to MinValue/MaxValue/PossibleValues and passes parameters in XML format to **Set Parameters** functions. If any parameter value is incorrect the **dialer** throws **DialerParametersException** which contains an enumeration of the **DialerParameterError** objects.

Note

The following parameters are obligatory, non-editable and hidden. The supplied XML file must always contain the configuration parameters listed below. These parameters have more simple structure – just name/value pairs. The parameter values depend on the particular **dialer** implementation:

SupportedPersonModes – comma separated list of agent task choice modes which the **dialer** supports. Possible task choice modes are: Automatic, Manual, CampaignAssignment:

```
<SupportedPersonModes> Automatic,Manual,CampaignAssignment</SupportedPersonModes>
```

IsReloginNeededOnCampaignChange – Does the **dialer** require the agent to re-login when the agent switches to another campaign?

<IsReLoginNeededOnCampaignChange>True</IsReLoginNeededOnCampaignChange>

IsHangUpSupported – Does the **dialer** support the hang-up command?

<IsHangUpSupported>True</IsHangUpSupported>

IsPauseOrResumePlaybackSupported – Does the **dialer** support pause/resume operation during voice fragment playback?

<IsPauseOrResumePlaybackSupported>False</IsPauseOrResumePlaybackSupported>

IsToggleAgentListensToPlaybackOrRespondentSupported – Does the **dialer** support switching voice source from the agent to file playback during voice fragment playback?

<IsToggleAgentListensToPlaybackOrRespondentSupported>True</IsToggleAgentListensToPlaybackOrRespondentSupported>

IsDynamicExtensionNumberAllowedForLocalAgents – Does the **dialer** allow entering interviewer extension number at local interviewer login?

<IsDynamicExtensionNumberAllowedForLocalAgents>False</IsDynamicExtensionNumberAllowedForLocalAgents>

IsDynamicExtensionNumberAllowedForRemoteAgents – Does the **dialer** allow entering interviewer extension number at remote interviewer login?

<IsDynamicExtensionNumberAllowedForRemoteAgents>False</IsDynamicExtensionNumberAllowedForRemoteAgents>

Dialer logs Access

Forsta CATI requires access to the dialer logs.

To that end **Forsta open dialer interface** uses a special **ILogger** interface with one function:

```
void Log (TraceEventType logLevel, string comment, string message)
```

Forsta CATI passes a reference to the logger via **Dialer driver assembly** main class constructor. See also [Instantiation](#).